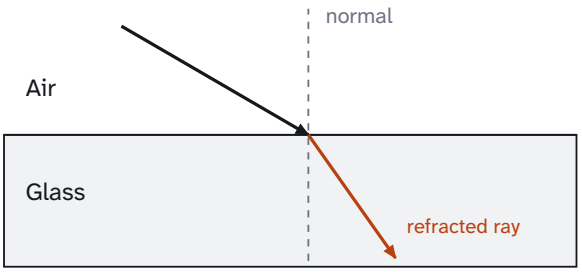
OW

Optics and Waves

Understand the idea. Build the method. Earn the marks.

FIG. 03 / OPTICS

A ray bends toward the normal in glass



At the boundary

Frequency: unchanged

Speed: decreases

Wavelength: decreases

Takeaway: measure angles from the normal, not from the glass surface.

Document	Optics and Waves
Level / lessons	3 / 12
Edition	November 2026 / Bench Sheet
Pages	66

Start here

Begin with one lesson and a pencil. Follow the stages below, then practise without notes.

1. Read the Idea and describe it in your own words before looking at the rule.
2. Check what every symbol means and when the rule applies.
3. Cover the worked steps. Predict the next step, then compare.
4. Use the working grid to attempt Your turn. Check the answers in the lesson review section.
5. Attempt the subject tests without notes. Use the explanations to identify the exact step you missed.

Read the page at a glance

Rule

$$v = \frac{d}{t}$$

A framed rule is something to select and apply, not just memorise. Here, average speed is distance divided by elapsed time.

Your turn

Blue marks a task for you. Write a prediction or calculation before checking.

Watch out

An orange note marks a common error. Example: do not confuse distance with displacement.

Before you begin

- Have a calculator, pencil and the units used in your course.
- Review basic rearrangement and graph reading when a lesson lists them as pre-requisites.
- Use the formula sheet and symbol glossary for revision; the same letter can have different meanings in different lessons.

Use the linked contents and alphabetical topic index to find a lesson. Marking points are study guidance, not an official marking scheme.

Your study route

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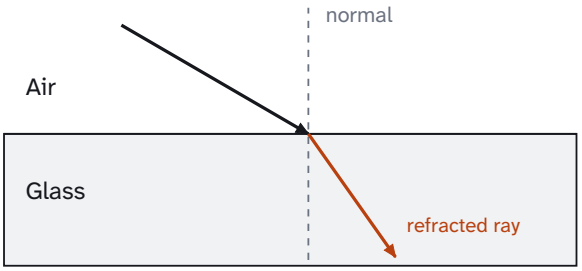
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Optics and Waves

FIG. 03 / OPTICS

A ray bends toward the normal in glass



At the boundary

Frequency: unchanged

Speed: decreases

Wavelength: decreases

Takeaway: measure angles from the normal, not from the glass surface.

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Subject map

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Wave language and simple harmonic motion p.6

02 Read a travelling wave without deriving it p.9

03 Wavefronts, spreading and group velocity p.13

04 Superposition, standing waves and beats p.16

05 Light, photons and the electromagnetic picture p.19

06 Reflection, refraction and total internal reflection p.22

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08 Optical systems, prisms and instruments p.29

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10 Polarization and Malus law p.36

11 Scattering, apparent depth and lens power p.39

12 Recognize advanced optics without long derivations p.42

Wave language and simple harmonic motion

OW / 01

- Convert an oscillation count into frequency and period.
- Relate displacement, speed and acceleration at key SHM positions.

You need: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

01 Idea

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

One complete oscillation returns the motion to the same position and direction. Period asks how long that return takes; frequency asks how many returns occur each second. They describe the same repeating motion from opposite viewpoints.

- More cycles within the same time means larger frequency.
- A larger frequency leaves less time for each cycle, so period decreases.

02 Rule and symbols

Rule

$$T = 1/f$$

$$f = N/t$$

$$a = -\omega^2 x$$

N number of complete cycles / dimensionless

t total elapsed time / s

f cycles per second / Hz

T time for one cycle / s

ω angular frequency / rad/s

03 Method

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

A counted number of complete oscillations over a measured time signals a frequency and period calculation.

04 Worked example

If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s.

01 Twenty complete oscillations in 5 s means 20/5 = 4 cycles per second, so f = 4 Hz.

02 One cycle takes 5/20 = 0.25 s, giving T = 0.25 s.

03 At an extreme the object must turn around, so its instantaneous speed is zero.

04 Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Watch out

The direction of particle motion is not necessarily the direction of wave travel.

Exam habit

Write f = N/t and T = t/N side by side to prevent interchanging hertz and seconds.

05 Your turn

A source makes 50 complete oscillations in 10 s. Find frequency and period.

If you get stuck, uncover one hint at a time:

1. Identify the complete cycle count and total time. Which ratio counts cycles per second?
2. Divide 50 cycles by 10 s for frequency.
3. Use total time divided by cycle count for period, or take the reciprocal of frequency. Check hertz and seconds respectively.

06 Check your understanding

At SHM equilibrium is acceleration magnitude maximum?

A source doubles its period. How does its frequency change?

1. It halves
2. It doubles
3. It stays unchanged

Answers and explanations: page 45.

Read a travelling wave without deriving it

OW / 02

- Read amplitude, wavelength and frequency from a travelling-wave expression.
- Use v = fλ and distinguish pattern motion from particle motion.

You need: Sine-wave shape; π; coefficients of x and t; SI units.

01 Idea

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

The source decides how often a wave repeats, while the medium sets its pattern speed under the stated conditions. Wavelength is the distance the pattern advances during one cycle. At fixed speed, more cycles each second mean each cycle occupies less distance.

- Frequency times wavelength gives distance travelled each second.
- In the same nondispersive medium, a frequency increase changes wavelength while speed stays fixed.

02 Rule and symbols

Rule

$$v = f\lambda$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$y = A\sin(kx - \omega t + \phi)$$

A amplitude or maximum displacement / m

k wave number / rad/m

ω angular frequency / rad/s

λ wavelength / m

f frequency / Hz

Use this when:

The example takes x and y in metres and t in seconds.

03 Method

1. Match the wave to y = A sin(kx - ωt + ϕ).
2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

Use the fixed-speed relationship when frequency changes but the same nondispersive medium is retained.

04 Worked example

For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

01 Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and ω = 20π rad/s.

02 Wavelength is 2π/(4π) = 0.50 m.

03 Frequency is 20π/(2π) = 10 Hz.

04 Pattern speed is 10 times 0.50 = 5 m/s; this is not the maximum transverse speed of an individual particle.

Watch out

The coefficient of time is angular frequency, not ordinary frequency.

Exam habit

Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

05 Your turn

A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

If you get stuck, uncover one hint at a time:

1. Use the original frequency and wavelength to establish speed first.
2. Retain that speed because the medium remains nondispersive and unchanged.
3. Divide the established speed by 12 Hz for the new wavelength; do not reuse the old wavelength with the new frequency.

06 Check your understanding

Is the coefficient 20π in the example the frequency in hertz?

A wave enters a medium where its speed is lower while frequency stays fixed. Its wavelength:

1. Must remain fixed
2. Becomes shorter
3. Becomes longer

Answers and explanations: page 45.

Phase means position within a cycle

A phase difference tells whether two oscillations are at the same or different stages of a cycle.

- One full cycle is 2π radians. Half a cycle is π radians.
- At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
- At one place, a full period brings the oscillation back to the same phase.
- These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.
- $\Delta\phi = 2\pi\dfrac{\Delta x}{\lambda}$ (phase-difference magnitude at one time)
- $\Delta\phi = 2\pi\dfrac{\Delta t}{T}$ (phase-change magnitude at one place)

Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

1. The separation is one quarter of a full spatial cycle.
2. Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

$\pi/2$ radians, or $90°$. They are neither in phase nor exactly opposite in phase.

Wavefronts, spreading and group velocity

OW / 03

- Predict intensity and amplitude changes for spherical spreading.
- Distinguish wavefront, ray, phase velocity and group velocity.

You need: Surface area grows with radius squared; square roots; amplitude versus intensity.

01 Idea

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

Picture fixed source power being shared over an expanding spherical shell. The larger shell carries the same total power when nothing is absorbed, but each square metre receives a smaller share. Geometric spreading can therefore reduce intensity without dissipating the total power.

- Sphere area grows with radius squared.
- With total power fixed, intensity decreases by the same area factor.

02 Rule and symbols

Rule

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

I energy-transfer rate per unit area / W/m^2

A wave amplitude / depends on disturbance

r distance from point source / m

Use this when:

The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

03 Method

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to $1/r^2$.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.
4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

An isotropic point source with negligible absorption permits inverse-square spreading; other geometries need their own model.

04 Worked example

For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

01 The same source power crosses a sphere around the source at each distance.

02 Doubling radius makes the sphere area four times as large.

03 The available power per square metre is therefore one quarter as large.

04 Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Watch out

The ideal plane-wave model does not have spherical inverse-square spreading.

Exam habit

Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

05 Your turn

At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

If you get stuck, uncover one hint at a time:

1. The new distance is three times the original. Ask how a sphere area changes when radius triples.
2. Square the distance factor for the inverse-square intensity change.
3. Apply that factor to 18 W/m^2 and retain intensity units; this is not a total-power calculation.

06 Check your understanding

An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Without absorption, which quantity stays the same on complete spheres farther from an ideal point source?

1. Intensity per square metre
2. Wave amplitude at each point
3. Total power crossing each sphere

Answers and explanations: page 45.

Superposition, standing waves and beats

OW / 04

- Calculate beat frequency from two nearby frequencies.
- Connect fixed-string harmonics with allowed wavelengths.

You need: Adding displacements; wavelength; frequency; fixed endpoints.

01 Idea

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

A string fixed at both ends must have nodes at its endpoints. Only patterns fitting whole numbers of loops between them can persist as normal modes. One loop spans half a wavelength, so a string length is not automatically a complete wavelength.

- Fixed endpoints restrict which wavelengths fit.
- At fixed wave speed, fitting more loops shortens wavelength and increases frequency.

02 Rule and symbols

Rule

$$f_{\text{beat}} = |f_1 - f_2|$$

$\lambda_n = 2L/n$ (string fixed at both ends)

f_{beat} number of loudness cycles per second / Hz

L vibrating string length / m

n harmonic number, 1, 2, 3, ... / dimensionless

λ_n wavelength of harmonic n / m

03 Method

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

For a string fixed at both ends, draw its node-and-loop pattern before selecting the harmonic wavelength.

04 Worked example

Two tones at 256 and 260 Hz produce 4 beats per second.

01 The tones have frequencies 256 Hz and 260 Hz.

02 Their frequency difference is 4 Hz, so there are four loudness beats each second.

03 For the separate string example, the fundamental has one loop between two end nodes.

04 That loop occupies half a wavelength, so $\lambda 1 = 2L$ and $f1 = v/(2L)$.

Watch out

A node is not a point where two travelling waves are individually absent.

Exam habit

A beat frequency is a difference, not the average of the two tone frequencies.

05 Your turn

A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

If you get stuck, uncover one hint at a time:

1. Draw the fixed endpoints and the single-loop fundamental.
2. The 1 m string contains half a wavelength. Find the full wavelength before using the 40 m/s speed.
3. Divide speed by wavelength for the fundamental frequency, then use the harmonic relationship for the second harmonic.

06 Check your understanding

For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Adjacent nodes on an ideal standing wave are separated by:

1. Half a wavelength
2. One wavelength
3. One quarter wavelength

Answers and explanations: page 45.

Light, photons and the electromagnetic picture

OW / 05

- Compare photon energy using frequency or vacuum wavelength.
- Distinguish beam brightness from energy per photon.

You need: Direct and inverse proportionality; wave frequency; energy transfer.

01 Idea

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

Photon energy belongs to an individual packet, whereas intensity describes energy arriving per area each second. Increasing frequency makes each packet more energetic. Making a fixed-colour beam brighter can instead increase the number of packets arriving, without changing the energy in each.

- Photon energy increases directly with frequency.
- Vacuum wavelength decreases as frequency increases because vacuum propagation speed is fixed.

02 Rule and symbols

Rule

$$E_\gamma = hf = hc/\lambda_0$$

$\lambda_0 =$ vacuum wavelength

$$c = 3.00 \times 10^8 \text{ m/s}$$

E_γ energy of one photon / J

h Planck constant / J s

f light frequency / Hz

λ vacuum wavelength / m

03 Method

1. Decide whether the question asks about one photon or the whole beam.
2. Use E_γ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.
4. For photoelectric emission compare energy per photon with the threshold requirement.

Use frequency ratios to compare photon energies and vacuum wavelengths without inserting unnecessary constants.

04 Worked example

Blue light has shorter wavelength than red light and therefore higher photon energy.

01 Blue light has a shorter vacuum wavelength than red light.

02 Since $E_\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon.

03 Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons.

04 Each red photon still has the same lower energy; brightness alone does not change its frequency.

Watch out

Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

Exam habit

Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

05 Your turn

Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

If you get stuck, uncover one hint at a time:

1. The supplied frequency ratio allows comparison without calculating Planck constant.
2. Photon energy follows the same ratio as frequency.
3. Vacuum wavelength follows the reciprocal frequency ratio. State which light has the larger energy and which has the shorter wavelength.

06 Check your understanding

If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Light crosses a stationary ordinary air-glass boundary. The transmitted photon energy:

1. Increases solely because the ray bends
2. Stays the same because frequency stays the same
3. Decreases because wavelength in glass is shorter

Answers and explanations: page 45.

Reflection, refraction and total internal reflection

OW / 06

- Track speed, frequency and wavelength across a refracting boundary.
- Recognize the two conditions for total internal reflection.

You need: Normal to a surface; sine; reciprocal refractive index.

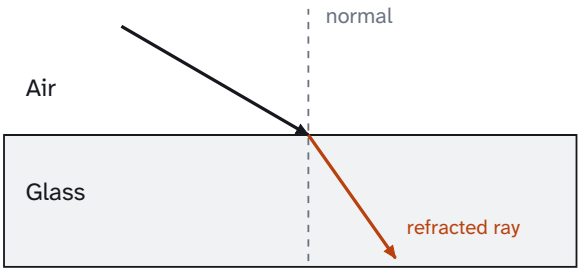
01 Idea

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.

A ray entering a lower-index medium bends away from the normal. As incidence angle increases, the transmitted direction approaches the boundary. At the critical angle the limiting transmitted ray runs along that boundary; beyond it the ordinary transmitted ray cannot propagate away into the second medium.

FIG. 03 / OPTICS

A ray bends toward the normal in glass



At the boundary

Frequency: unchanged

Speed: decreases

Wavelength: decreases

Takeaway: measure angles from the normal, not from the glass surface.

Figure 03. At oblique incidence from air into glass, the ray bends toward the normal. At a stationary boundary its frequency is unchanged, while speed and wavelength decrease.

- Only higher-to-lower index travel can reach the critical-angle condition.
- Total internal reflection requires incidence above, rather than merely equal to, the critical angle.

02 Rule and symbols

Rule

$$\theta_i = \theta_r$$

$$n = c/v$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_c = n_2/n_1$$

n refractive index / dimensionless

c vacuum light speed / m/s

v light speed in medium / m/s

θ_c critical incidence angle / degrees or radians

Use this when:

The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

03 Method

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

The words critical angle signal a check of travel direction followed by the lower-to-higher index sine ratio.

04 Worked example

From air into glass with index 1.5, light slows to $2 \times 10^8\ \mathrm{m/s}$.

01 Air is approximated by n = 1 and glass has n = 1.5.

02 In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s.

03 The frequency is set by the source and remains the same across the boundary.

04 Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Watch out

The critical angle is an incidence angle in the higher-index medium.

Exam habit

At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

05 Your turn

Find the critical angle for glass of index 1.5 into air of index 1.0.

If you get stuck, uncover one hint at a time:

1. Confirm that light travels from glass index 1.5 into air index 1.0.
2. Divide the lower index by the higher index to obtain the critical-angle sine.
3. Take inverse sine and report an angle from the normal. Check calculator degree mode if using degrees.

06 Check your understanding

Can light travelling from air into glass undergo total internal reflection at that interface?

At exactly the critical angle, the refracted ray travels:

1. Along the normal

2. Back along the incident ray
3. Along the boundary

Answers and explanations: page 45.

Mirrors, lenses and image signs

OW / 07

- Use the stated image-distance sign convention consistently.
- Infer image reality, orientation and relative size from di and m.

You need: Reciprocals; positive and negative numbers; a simple ray sketch.

01 Idea

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

A lens equation locates where rays meet or appear to originate. Magnification then compares image height with object height, including orientation. A negative magnification records an inverted image; it does not mean the physical image has a negative length.

- A converging lens with an object at twice focal length gives a real image at twice focal length on the other side.
- Magnitude of magnification gives relative size, while its sign gives orientation.

02 Rule and symbols

Rule

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

f signed focal length / cm or m consistently

d_o object distance / same length unit as f

d_i signed image distance / same length unit as f

m transverse magnification / dimensionless

Use this when:

Use the lesson convention: real object and real image distances positive; converging focal length positive.

03 Method

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

Use one consistent sign convention when focal length and object distance are given, then interpret the numerical result.

04 Worked example

For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

01 A converging lens has f = +10 cm and the real object has do = +30 cm.

02 Then 1/di = 1/10 - 1/30 = 1/15 per cm, so di = +15 cm.

03 Magnification is -15/30 = -0.5.

04 Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Watch out

Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

Exam habit

After subtraction, invert 1/di to get di; include all three requested image descriptions.

05 Your turn

A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

If you get stuck, uncover one hint at a time:

1. Assign positive focal length to the converging lens and positive distance to the real object.
2. Subtract the reciprocal of 24 cm from the reciprocal of 12 cm, then invert for image distance.
3. Calculate signed magnification and interpret distance sign, orientation and size separately.

06 Check your understanding

What does m = +2 tell you about an image?

Why can a real image be projected onto a screen?

1. Actual rays meet at its position
2. Its magnification must be positive
3. All images lie on the lens surface

Answers and explanations: page 45.

Optical systems, prisms and instruments

OW / 08

- Explain the different roles of objective and eyepiece.
- Calculate telescope angular magnification for normal adjustment.

You need: Focal length; image formation; ratios of lengths.

01 Idea

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

A telescope objective forms an intermediate image of a distant scene, and an eyepiece enlarges the angle under which it is viewed. Angular magnification compares viewing angles, so it is different from the height ratio for one ordinary lens image.

- For normal adjustment, a shorter eyepiece focal length gives larger angular magnification with the same objective.
- A focal-length ratio is dimensionless when both lengths use the same unit.

02 Rule and symbols

Rule

$$m_{\text{total}} = m_1 m_2$$

$$|M_{\text{telescope}}| = f_o / f_e \text{ (normal adjustment)}$$

f_o objective focal length / cm or m consistently

f_e eyepiece focal length / same length unit as fo

M angular magnification / dimensionless

m_1, m_2 transverse magnifications of successive stages / dimensionless

Use this when:

The telescope is in normal adjustment, giving the final image at infinity.

03 Method

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

Use the focal-length ratio for an astronomical telescope in normal adjustment, not for an arbitrary two-lens system.

04 Worked example

An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20.

01 The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm.

02 Their ratio is 80/4 = 20, so the angular magnification magnitude is 20.

03 An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image.

04 The ratio needs no length conversion because both focal lengths use the same unit.

Watch out

Linear image magnification and angular magnification are different quantities.

Exam habit

Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

05 Your turn

A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

If you get stuck, uncover one hint at a time:

1. Identify the objective focal length and eyepiece focal length.
2. Divide 100 cm by 5 cm for initial magnification magnitude.
3. Replace only the denominator with 10 cm and compare ratios. Keep image orientation separate from the requested magnitude.

06 Check your understanding

Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Which eyepiece gives greater magnification with the same objective in normal adjustment?

1. Both give equal magnification
2. The shorter-focal-length eyepiece
3. The longer-focal-length eyepiece

Answers and explanations: page 45.

Interference and diffraction

OW / 09

- Predict how slit geometry changes an interference or diffraction pattern.
- Distinguish double-slit bright fringes from single-slit dark fringes.

You need: Wavelength; coherent sources; nanometres and millimetres; small angles.

01 Idea

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

Coherent slits create bright and dark directions through path differences between overlapping waves. A screen turns those angles into distances. Moving the screen farther away enlarges the visible spacing even though the slit geometry and bright angular directions remain unchanged.

- Longer wavelength increases the angular change needed between neighbouring bright fringes.
- A more distant screen maps the same angular separation to a larger physical distance.

02 Rule and symbols

Rule

$$\Delta y = \lambda L / d$$

$d \sin\theta = m\lambda$ (double-slit bright)

$a \sin\theta = m\lambda$ (single-slit dark, $m \neq 0$)

Δy double-slit fringe spacing / m

L slit-to-screen distance / m

d separation of double slits / m

a single-slit width / m

λ wavelength / m

Use this when:

Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

03 Method

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use λL/d with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

For two coherent narrow slits and a distant screen, use wavelength, slit separation and screen distance to find spacing.

04 Worked example

Doubling screen distance doubles double-slit fringe spacing.

01 Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed.

02 Replacing L by 2L therefore doubles fringe spacing.

03 In the separate single-slit comparison, the first dark direction obeys sin θ = λ/a.

04 Halving a makes that angle larger, so the bright central maximum broadens.

Watch out

The double-slit bright condition is not the single-slit bright condition.

Exam habit

Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

05 Your turn

In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

If you get stuck, uncover one hint at a time:

1. Distinguish slit separation from slit width; this is a double-slit spacing question.
2. Convert 500 nm and 0.50 mm to metres before combining them with 2 m.
3. Multiply wavelength by screen distance and divide by slit separation. Convert the final spacing to millimetres if helpful.

06 Check your understanding

If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Only the double-slit screen distance doubles. The bright-fringe angular directions:

1. Double along with screen spacing
2. All become zero
3. Stay unchanged while screen spacing grows

Answers and explanations: page 45.

Recognize gratings, Michelson paths and thin films

All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

- A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
- A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
- A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
- For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.
- $\Delta\text{path} = 2\Delta x$ (one Michelson mirror moves)

In a Michelson interferometer using $500\,\text{nm}$ light in air, one mirror moves $250\,\text{nm}$ along its arm. What path change and fringe shift result?

1. Round-trip path change: $2 \times 250 = 500\,\text{nm}$.
2. This equals one wavelength.

The path changes by $500\,\text{nm}$, so one fringe passes a fixed observation point.

Polarization and Malus law

OW / 10

- Apply the initial half-intensity rule to unpolarized light.
- Apply Malus law using the angle between polarization and analyzer axes.

You need: Cosine of common angles; intensity versus field amplitude; sequential fractions.

01 Idea

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

The first ideal polarizer turns unpolarized light into linearly polarized light while transmitting half its intensity. The analyzer receives that output. It then passes a fraction determined by the angle to the already established polarization, so the first and second stages use different rules.

- The half-intensity rule applies to initially unpolarized input.
- For linearly polarized input, the analyzer transmits a squared-cosine fraction.

02 Rule and symbols

Rule

$$I = I_0 \cos^2 \theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

I_0 intensity incident on the analyzer / W/m^2 or stated units

I intensity transmitted by analyzer / same as I0

θ angle between incident polarization and analyzer axis / degrees or radians

Use this when:

The calculation assumes ideal polarizers and initially unpolarized illumination.

03 Method

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by $\cos^2 \theta$, where θ is between the two relevant axes.

For successive polarizers, identify input polarization and process each element in order.

04 Worked example

Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees.

01 The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units.

02 The analyzer axis is at 60° to the first polarizer axis.

03 Cos 60° is 0.5 and its square is 0.25.

04 The analyzer therefore transmits 50 times 0.25 = 12.5 units, one eighth of the original intensity.

Watch out

Do not halve already polarized intensity again before applying Malus law.

Exam habit

Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

05 Your turn

Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

If you get stuck, uncover one hint at a time:

1. Start with the unpolarized 80 W/m^2 and apply the first-polarizer rule.
2. Use that transmitted intensity as the analyzer input.
3. Multiply by the squared cosine of 45 degrees. Keep the two stages visible rather than applying an extra half factor.

06 Check your understanding

Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

A second ideal polarizer is aligned with the first. What fraction of the first output passes it?

1. All of it
2. Half of it
3. None of it

Answers and explanations: page 45.

Scattering, apparent depth and lens power

OW / 11

- Calculate apparent depth near normal viewing and signed lens power.
- Explain why small-particle scattering favours shorter wavelengths.

You need: Refraction; inverse relationships; centimetres-to-metres conversion.

01 Idea

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

Lens power describes how strongly a lens converges or diverges rays. A shorter focal-length magnitude means stronger bending and greater power magnitude. The negative sign of a diverging lens communicates the direction of its optical effect, rather than the physical size of the lens.

- Reciprocals make smaller focal-length magnitudes correspond to larger power magnitudes.
- Dioptres use metres in the focal-length reciprocal, so centimetres must be converted first.

02 Rule and symbols

Rule

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

d_{real} actual depth / m

d_{apparent} apparent depth / m

n liquid refractive index relative to air approximation / dimensionless

P lens optical power / D = m^{-1}

f signed focal length / m

Use this when:

The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

03 Method

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.
2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

A supplied focal length and a request for optical power call for a signed reciprocal in metre units.

04 Worked example

A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal.

01 For the pool, near-normal apparent depth is 2/1.33 ≈ 1.50 m, so the bottom appears shallower.

02 The index does not change the real 2 m depth; it changes the directions of the emerging rays.

03 For the separate converging lens, f = +0.25 m.

04 Its power is 1/0.25 = +4 m^{-1}, written +4 dioptres.

Watch out

A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

Exam habit

A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

05 Your turn

A diverging lens has focal length -50 cm. Find its optical power.

If you get stuck, uncover one hint at a time:

1. Retain the negative focal-length sign because the lens diverges.
2. Convert -50 cm into metres.
3. Take the signed reciprocal and report dioptres. A positive result would conflict with the stated lens type.

06 Check your understanding

A diverging lens has focal length -50 cm. What is its power?

Which diverging lens has greater power magnitude: focal length -0.25 m or -0.50 m?

1. They match because both diverge
2. The -0.25 m lens
3. The -0.50 m lens

Answers and explanations: page 45.

Recognize advanced optics without long derivations

OW / 12

- Recognize what a quarter-wave plate changes about polarized light.
- Use Brewster angle and plane-wave relations at recognition level.

You need: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

01 Idea

The wave equation describes a disturbance propagating in space and time. Complex notation is a bookkeeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

Brewster incidence is a special transparent-interface geometry. In ordinary transparent nonmagnetic media, the reflected light at that angle lacks the component polarized in the incidence plane. This concerns polarization of the reflected component, not complete reflection of all incoming energy.

- The refractive-index ratio sets the tangent of the Brewster angle from the normal.
- Brewster incidence and total internal reflection describe different conditions and effects.

02 Rule and symbols

Rule

$$\tan\theta_B = n_2/n_1$$

$E = cB$ (plane wave in vacuum)

θ_B Brewster incidence angle from the normal / degrees or radians

n_1, n_2 incident and transmitted medium indices / dimensionless

E plane-wave electric-field magnitude / V/m

B plane-wave magnetic-field magnitude / T

Use this when:

The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

03 Method

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

Use Brewster angle when an interface question links refractive indices and linearly polarized reflection.

04 Worked example

A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components.

01 Resolve suitable incident linear polarization along the plate principal axes.

02 Equal component amplitudes are required for the stated circular-polarization result.

03 A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay.

04 Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Watch out

These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

Exam habit

For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

05 Your turn

For light from air into glass of index 1.5, estimate Brewster angle.

If you get stuck, uncover one hint at a time:

1. Choose the Brewster tangent relation, not the critical-angle sine relation.
2. Place glass index 1.5 over air index approximately 1.0.
3. Take inverse tangent of that ratio and measure the result from the normal. Check degree mode if reporting degrees.

06 Check your understanding

Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

For unpolarized light at Brewster incidence in ordinary transparent nonmagnetic media:

1. All incident light must be reflected
2. Total internal reflection must occur
3. The reflected light is linearly polarized

Answers and explanations: page 45.

Lesson review and answers

Wave language and simple harmonic motion

Return to lesson: page 6.

Your turn: No. Displacement and restoring acceleration are zero; speed is maximum.

Concept check: 1. Correct. Twice the time per cycle gives half as many cycles each second.

Written method: Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Read a travelling wave without deriving it

Return to lesson: page 9.

Your turn: No. It is angular frequency; divide it by 2π to get 10 Hz.

Concept check: 2. Correct. At fixed frequency, lower speed means less distance travelled per cycle.

Written method: Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Wavefronts, spreading and group velocity

Return to lesson: page 13.

Your turn: 2 W/m^2: divide by 3^2 = 9.

Concept check: 3. Correct. The unchanged source power is shared over each complete sphere.

Written method: Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Superposition, standing waves and beats

Return to lesson: page 16.

Your turn: It is twice the fundamental frequency in the ideal uniform-string model.

Concept check: 1. Correct. One loop between neighbouring nodes occupies half a wavelength.

Written method: Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Light, photons and the electromagnetic picture

Return to lesson: page 19.

Your turn: A has twice the photon energy and half the vacuum wavelength.

Concept check: 2. Correct. Unchanged frequency gives unchanged photon energy.

Written method: A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.

Reflection, refraction and total internal reflection

Return to lesson: page 22.

Your turn: No. Total internal reflection requires travel from higher to lower refractive index.

Concept check: 3. Correct. The transmitted angle from the normal is 90 degrees.

Written method: Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Mirrors, lenses and image signs

Return to lesson: page 26.

Your turn: It is upright and twice the object height; the sign convention and lens setup determine its image position.

Concept check: 1. Correct. A screen intercepts the rays where they form the image.

Written method: 1/di = 1/12 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Optical systems, prisms and instruments

Return to lesson: page 29.

Your turn: It halves the magnification magnitude.

Concept check: 2. Correct. The eyepiece focal length is the denominator of the ratio.

Written method: Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Interference and diffraction

Return to lesson: page 32.

Your turn: It halves.

Concept check: 3. Correct. Wavelength and slit separation set directions; screen distance converts direction to position.

Written method: Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.

Polarization and Malus law

Return to lesson: page 36.

Your turn: Zero intensity in the ideal model, because $\cos^2 90° = 0$.

Concept check: 1. Correct. The input is already aligned, so the squared cosine of zero is one.

Written method: First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Scattering, apparent depth and lens power

Return to lesson: page 39.

Your turn: -2 D, since -50 cm = -0.50 m.

Concept check: 2. Correct. The smaller focal-length magnitude has the larger reciprocal magnitude.

Written method: Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.

Recognize advanced optics without long derivations

Return to lesson: page 42.

Your turn: No. The two orthogonal components must also have equal amplitudes.

Concept check: 3. Correct. The reflected component lacks one linear polarization component.

Written method: Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Optics and Waves / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which wave can propagate through vacuum?

   A. A water wave

   B. Light

   C. A stretched-string wave

   D. Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   A. Half as large

   B. Unchanged

   C. Four times as large

   D. Twice as large

3. Group velocity describes the motion of:

   A. The wave-packet envelope

   B. The source itself

   C. One particle only

   D. A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   A. 1320 Hz

   B. 883 Hz

   C. 3 Hz

   D. 441.5 Hz

5. At the same medium, which has greater photon energy?

   A. Lower-frequency light

   B. Both always equal

   C. Higher-frequency light

D. Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

A. Ray direction in every case

B. Speed

C. Wavelength

D. Frequency

Part II / Show your reasoning

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. An image with magnification -2 is:

   A. Upright and smaller

   B. Inverted and twice as tall

   C. Virtual in every case

   D. Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   A. A longer focal length than the eyepiece

   B. Zero focal length

   C. The same role as a polarizer

   D. No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   A. Becomes negative

   B. Increases

   C. Stays fixed

   D. Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   A. One quarter

   B. Zero

   C. All its intensity

   D. Half

5. The unit of optical lens power is:

   A. Hertz

   B. Watt

C. Dioptre

D. Newton

6. A quarter-wave plate acts primarily by changing the relative:

A. Charge of light

B. Speed of the source

C. Mass of photons

D. Phase of orthogonal components

Part II / Show your reasoning

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. Light is electromagnetic and does not require a material medium.

Written solution: Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Return to lesson: page 6.

Question 2

MCQ: A. Wavelength equals speed divided by frequency.

Written solution: Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Return to lesson: page 9.

Question 3

MCQ: A. The group is the envelope formed by combining nearby frequencies.

Written solution: Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Return to lesson: page 13.

Question 4

MCQ: C. Beat frequency is the absolute difference.

Written solution: Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Return to lesson: page 16.

Question 5

MCQ: C. Photon energy is proportional to frequency.

Written solution: A has twice the photon energy and half the vacuum wavelength.

Two marks for each relationship.

Return to lesson: page 19.

Question 6

MCQ: D. The boundary does not change the frequency of the incident oscillation.

Written solution: Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Return to lesson: page 22.

Optics and Waves / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. The minus sign indicates inversion and magnitude 2 gives the size ratio.

Written solution: 1/di = 1/12 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Return to lesson: page 26.

Question 2

MCQ: A. The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.

Written solution: Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Return to lesson: page 29.

Question 3

MCQ: D. Fringe spacing is inversely proportional to slit separation.

Written solution: Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.

Return to lesson: page 32.

Question 4

MCQ: B. Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.

Written solution: First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Return to lesson: page 36.

Question 5

MCQ: C. Lens power is inverse focal length in metres, expressed in dioptres.

Written solution: Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.

Return to lesson: page 39.

Question 6

MCQ: D. A controlled phase delay changes the polarization state.

Written solution: Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Return to lesson: page 42.

Formula sheet

Wave language and simple harmonic motion p.6

Rule

$$T = 1/f$$
$$f = N/t$$
$$a = -\omega^2 x$$

Read a travelling wave without deriving it p.9

Rule

$$v = f\lambda$$
$$k = 2\pi/\lambda$$
$$\omega = 2\pi f$$
$$y = A\sin(kx - \omega t + \phi)$$

Wavefronts, spreading and group velocity p.13

Rule

$$I \propto A^2$$
$$I_{\text{spherical}} \propto 1/r^2$$

Superposition, standing waves and beats p.16

Rule

$$f_{\text{beat}} = |f_1 - f_2|$$
$\lambda_n = 2L/n$ (string fixed at both ends)

Light, photons and the electromagnetic picture p.19

Rule

$$E_\gamma = hf = hc/\lambda_0$$
$\lambda_0 =$ vacuum wavelength
$$c = 3.00 \times 10^8 \text{ m/s}$$

Reflection, refraction and total internal reflection p.22

Rule

$$\theta_i = \theta_r$$
$$n = c/v$$
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\sin\theta_c = n_2/n_1$$

Mirrors, lenses and image signs p.26

Rule

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

Optical systems, prisms and instruments p.29

Rule

$$m_{\text{total}} = m_1 m_2$$

$|M_{\text{telescope}}| = f_o/f_e$ (normal adjustment)

Interference and diffraction p.32

Rule

$$\Delta y = \lambda L/d$$

$d \sin\theta = m\lambda$ (double-slit bright)

$a \sin\theta = m\lambda$ (single-slit dark, $m \neq 0$)

Polarization and Malus law p.36

Rule

$$I = I_0 \cos^2\theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

Scattering, apparent depth and lens power p.39

Rule

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$P_{\text{lens}} = 1/f$ (f in metres)

Recognize advanced optics without long derivations p.42

Rule

$$\tan\theta_B = n_2/n_1$$

$E = cB$ (plane wave in vacuum)

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Wave language and simple harmonic motion

N number of complete cycles / dimensionless

t total elapsed time / s

f cycles per second / Hz

T time for one cycle / s

ω angular frequency / rad/s

Read a travelling wave without deriving it

A amplitude or maximum displacement / m

k wave number / rad/m

ω angular frequency / rad/s

λ wavelength / m

f frequency / Hz

Wavefronts, spreading and group velocity

I energy-transfer rate per unit area / W/m^2

A wave amplitude / depends on disturbance

r distance from point source / m

Superposition, standing waves and beats

f_{beat} number of loudness cycles per second / Hz

L vibrating string length / m

n harmonic number, 1, 2, 3, … / dimensionless

λ_n wavelength of harmonic n / m

Light, photons and the electromagnetic picture

E_γ energy of one photon / J

h Planck constant / J s

f light frequency / Hz

λ vacuum wavelength / m

Reflection, refraction and total internal reflection

n refractive index / dimensionless

c vacuum light speed / m/s

v light speed in medium / m/s

θ_c critical incidence angle / degrees or radians

Mirrors, lenses and image signs

f signed focal length / cm or m consistently

d_o object distance / same length unit as f

d_i signed image distance / same length unit as f

m transverse magnification / dimensionless

Optical systems, prisms and instruments

f_o objective focal length / cm or m consistently

f_e eyepiece focal length / same length unit as fo

M angular magnification / dimensionless

m_1, m_2 transverse magnifications of successive stages / dimensionless

Interference and diffraction

Δy double-slit fringe spacing / m

L slit-to-screen distance / m

d separation of double slits / m

a single-slit width / m

λ wavelength / m

Polarization and Malus law

I_0 intensity incident on the analyzer / W/m^2 or stated units

I intensity transmitted by analyzer / same as I0

θ angle between incident polarization and analyzer axis / degrees or radians

Scattering, apparent depth and lens power

d_{real} actual depth / m

d_{apparent} apparent depth / m

n liquid refractive index relative to air approximation / dimensionless

P lens optical power / D = m^{-1}

f signed focal length / m

Recognize advanced optics without long derivations

θ_B Brewster incidence angle from the normal / degrees or radians

n_1, n_2 incident and transmitted medium indices / dimensionless

E plane-wave electric-field magnitude / V/m

B plane-wave magnetic-field magnitude / T

Source and coverage notes

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Topic index

Find a lesson alphabetically. For individual symbols, use the subject symbol glossaries listed in the contents.

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